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A Moore method calculus II course

William S. Mahavier

Emory University

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Introduction

A few years ago a college committee changed the Uniform Requirements for a degree at Emory. Following a trend that seems to be sweeping the country they announced that each student at Emory must take a freshman seminar and an upper level seminar style course in order to graduate. The definition of a seminar was rather loose but the idea was that the students would be active participants in the course and the enrollment would be limited to 15 students. I received my PhD degree under the supervision of R. L. Moore and whenever a syllabus permitted it I have taught my courses using what is called the Moore method. That means different things to different people, but probably to almost everyone it requires that the students be active participants in the course. I proposed and have taught for several years a freshman seminar course for our freshmen who made a 4 or 5 on the AB version of the Calculus AP exam. In the last section I will include information on how the course has been evaluated including student evaluations, comments and some tracking of students from the course. In this article I will describe the way I have conducted the course and provide a set of problems that I have presented in the past. The reader should realize that this is a work in progress and that I have altered the way I conducted the course each time I have taught it, often as a result of suggestions from students. I have also changed the problem sequence each semester as I better learn what the students are able to do. It is my hope that others will try their hand at such a course. I also believe that one does not need such a carefully selected group of students and that with the proper problem sequence and appropriate help when needed that a similar course could be given to any Calculus II course. The only limitation being that the class size would probably have to be limited to perhaps 20 at most. One would have to adjust the problem sequence to the course and to the syllabus to be covered.

I would like to emphasize that it is not my intent to tell anyone how to teach. I hope to merely convey what I do in my classes. Another instructor would surely want to vary the amount of lecturing and alter the problem sequence to suit their own teaching style. I do hope this monograph will provide a start and some incentive for others to try a similar course.

An observer's view

In the Fall of 2001, supported by a grant from the Educational Advancement Foundation, Dr. Stallmann of Augusta State University spent a semester at Emory and observed my Calculus seminar and another seminar style course I taught in analysis. At the end of that semester he wrote a description of the course from his perspective and, with his permission, I include below part of that document. The following is quoted from Professor Stallmann's write up.

What I did not at all appreciate until my experience watching Bill this past semester was how very important the careful use of language is. This has perhaps been the single most important lesson that I learned. Using good language and good notation can greatly facilitate the learning of mathematics, particularly for those who have to struggle with it. I think this point is often overlooked by math teachers. We use language whose ambiguities and inaccuracies we have somehow been able to overcome.

Another aspect of Bill's notes that I have come to appreciate is the fact that they are essentially self contained. The student is not penalized because of something that is missing from his background. There is nothing like the sense that you don't know something that you should know to keep you quiet and reluctant to contribute.

A final essential feature of Bill's notes is their focus. There is no attempt to be comprehensive, to cover a huge amount of material. This gives the student a much better chance to develop understanding.

In addition to getting a full appreciation for how and why the notes are designed the way they are there are many other features of Bill's teaching that only direct observation can properly convey. The first few weeks of class Bill makes a very concerted effort to get everyone to the board, to make everyone feel comfortable enough and have enough success to want to present again. In short he gets the students hooked. Once this is accomplished the students will work hard and fight to go to the board. This part of the course requires an inordinate amount of patience, a great respect for the students, and above all a great deal of experience. Having watched how Bill handled the first couple weeks of his course will have enormous benefits to my own attempts at teaching in this way. If I had not observed Bill I think I would have been ready to show the students too much early on, to interfere, and probably to be too critical.

Another feature of Bill's version of the Moore method that impressed me is the great extent that Bill does provide help to the students when they need it. My impression of students left to their own devices to figure things out is far from what Bill and R. L. Moore had in mind. His help includes well placed and sometimes quite subtle hints, frequent one on one office consultations, and also the occasional lecture.

There are numerous other ways in which my impression of the Moore Method has *changed for the better, some of which I summarize below. The most important change, however, is that I now firmly believe that it does work. The accomplishments of the students in his classes and the very positive comments expressed by them about the course at the end of the semester is enough to convince me of that.*

New impressions of the Moore Method

1. Constructing a good problem sequence requires a great deal of care and thought and should:

> *a) allow students to get started right away and so that even weaker students will have early successes.*

> *b) use language and notation that is straight forward, avoids misleading the students. The language should help and not hinder understanding.*

> *c) keep everything as self contained as possible. The number of axioms, theorems, or techniques should be kept to a minimum, especially those that depend on knowledge from previous courses.*

2. A great deal of patience should be exercised particularly at the beginning. The students are not only grappling with difficult mathematical concepts but also getting used to a new way of learning. They will need a great deal of encouragement. Once they get hooked the job will be much easier.

3. Students need to be convinced as Bill has said many times, "That you are part of the solution and and not part of the problem." You need to show them that they can solve difficult problems and that it can be fun.

4. The students will need a great deal of help especially in the beginning. It should be given in a way that is best suited to the particular student.

5. An occasional well placed lecture can be quite beneficial to clarify something that most of the class is having trouble with.

8. Grading should be consistent with the goals of the course. It will necessarily be subjective. The students should know this up front. For some students good grades will be an encouragement, others might need a bad grade to get them to work.

9. Written assignments with opportunities for follow up corrections are a good way to keep everyone engaged whether they are presenting at the board or not.

The syllabus

On the first day I hand out a sheet which contains the usual information as to my name, office hours, email address since I encourage students to use email to contact me for help or to set up an appointment. I also hand out something about the course and how it will be graded. The following is typical of what I hand out at the beginning of the period. I give them time to read the sheet and ask questions.

1. WHAT ARE THE GOALS OF THE COURSE?

I have several goals in this course. Probably my primary goal is to convince you that you can completely understand everything that is done in calculus and that you are capable of deriving formulas instead of having them given to you to use. Almost equally important I hope to convince you that working problems or discovering facts or proving theorems are pretty much all the same and is a lot of fun! You will learn to express your ideas carefully and precisely both verbally and in writing and this will be valuable to you in many ways that have nothing to do with mathematics. You will learn the importance of understanding and using definitions and axioms in solving problems.

2. HOW WILL THE COURSE BE CONDUCTED?

This will depend on the class but my usual policy is to hand out sheets on a regular basis. We will not use a text. The sheets will have some discussion and some definitions and problems. I will expect you to carefully read my handouts and to work on all the problems. It will be my job to give you problems that are hard enough to be challenging but not so hard as to be discouraging. You will never be required to present a solution to a problem in class until you have had time at home to work on it. I will almost never lecture except perhaps in answer to a question. This is a seminar style course so you will be expected to do the talking in class instead of me. You will sometimes lecture and I hope there will often be class discussions of the problems being worked. I will at first ask for volunteers to go to the board and show what they have done trying to solve a problem. You do not need to solve a problem before going to the board. All you have to do is show me and the class what you have done. If I, or someone in the class, has an idea that might help you finish the problem, and you want help, we can offer it. You can then work more on the problem, with or without help, in class or at home later, or you can give up and let someone else try to solve it. It will be up to you as it is "your problem" as long as you want to work on it. Any time you go to the board that will help your "class performance" grade whether you get the problem or not. Of course if you do finish the problem that will help even more. Going to the board will never hurt your class performance grade. I will keep track of who has been to the board and how well they have done and after a few days I will begin to call on those who have been to the board least. This will be my policy all semester so that those whose class performance grade is lowest will get first chance to raise their score. This score will necessarily be subjective but you can ask me in my office or via email at any time what my opinion of your class performance grade is at that time.

3. HOW WILL YOUR GRADE BE DETERMINED?

I will give two examinations. An in class mid-term exam sometime after the course is half over and a final exam at the regularly scheduled time. Your grade on these exams will count about 30% of your final grade. Probably I will count the midterm about 12% and the final about 18%. To do well on the test you should at least try every one of the problems. If you work on a problem at home, you are more likely to understand what is done on that problem in class and this will be good preparation for our exams.

We will select one day each week on which homework will be due. If that day is, for example Monday, then you will be required to turn in each Monday a written solution to one of our problems using complete and correct English sentences. You may pick the problem you want to write up. You should pick one that has been done in class and you think you understand. If you don't like the grade you get, then you can re-write the same problem and turn it in the following Monday and I will count only the best of the two grades. But you must also turn in a new problem on that day. However, on your 2^{nd} write up, the highest grade you can receive is a B+. You must also turn in a new problem each week whether you do any rewriting or not.

I will not accept late work even if you miss class. If you have to miss class then you can have someone bring your work to me. I will grade your paper only if it is turned in on the day the class meets. I realize that many of you will have valid reasons for missing an assignment so at the end of the year I will drop one or two of your lowest homework grades. I will count your homework grade about 35%. I do not object to your working together on problems or discussing problems together before or after they are done in class. I do ask that once you decide to write up a problem to turn in that you do not discuss your write up work with anyone except me until it has been graded and given back.

Finally your class participation grade will count about 35% of your final grade. In addition to what I said before about your work at the board I will also count comments and discussions from your seat as part of your class participation grade. Be aware that class attendance is very important. Much of your grade depends on class participation and when you miss a class you not only do not contribute on that day but you are not likely to be able to contribute at the next class meeting. If you must miss class you should try to see me in my office to find out what happened before the next class meeting. And it is your responsibility to see to it that you get any handouts or homework returned that you miss because of an absence.

Let me finally emphasize that mathematics is not a spectator sport. You learn very little mathematics watching someone do mathematics. It is much like trying to learn to ride a bicycle by watching someone ride a bicycle. Actually you would probably learn more about riding a bicycle by watching someone who is trying to learn how to ride than you would by watching the Tour de France. I think you will learn more by watching your fellow students try to solve a problem than you would learn by watching me solve the problem. Your should learn from their mistakes or details omitted. I hope you will learn some logic, how to think clearly and some calculus and have fun doing it.

Chapter 1

The first problems

- The comments, like this one which are indented and preceded by a • are comments about how I conduct the class. The text which is not indented are typical of the sheets that I hand out to the students.

  On the first day, after the usual introductions and a brief explanation of the course as described in the preceding section, I begin by indicating that I plan to have them develop the properties of the logarithm and exponential functions as defined by a definite integral. I emphasize that they are not to use any of the properties of the logarithm or exponential functions that they learned in high school. I also tell them that they do not need to copy any of the definitions or problems as I state them as I will hand out a sheet at the end of class which will contain this information.

  On a daily basis I decide which problems I will hand out next, especially at first as I try to get to know the students and their abilities. Later in the course I will had out more problems at a time.

  Also, I have found that these students have no idea how to work with inequalities so I think that the next time I teach this course I will include some problems about inequalities before they are needed in the study of convergent sequences or series.

  As an example, they have trouble explaining why $b-a$ is positive if and only if $b > a$.

Definition 1. *If a and b are numbers and $a < b$, then by the* **interval** *from a to b is meant the set of all numbers x such that $a \leq x \leq b$.*

Notation 1. *If a and b are numbers and $a < b$ then we will use $[\mathbf{a},\mathbf{b}]$ to represent the interval from a to b. Also if we say to let $[a,b]$ be an interval, we will mean that a is a number, b is a number, and $a < b$.*

- Here I would emphasize that an interval includes its endpoints.

- I want to define and get them to use Riemann sums. I find that they are not likely to remember much about Riemann sums except that there were formulas for them. Some will remember formulas for the left and right endpoint Riemann sums, mid-point Riemann sums and the trapezoid rule. They are not likely to have any geometric representation for these sums. Thus I start very slowly.

I first describe geometrically a lower Riemann sum and an upper Riemann sum as the sum of the areas of some rectangles which I will sketch for an increasing or a decreasing function.

The reader should keep in mind that the notes included above and below will be on the first sheet that I will hand out at the end of the first period but they have not seen them. So as I define the function h by $h(x) = \frac{1}{x}$, I will graph it, explaining that I call it h because it is part of a hyperbola. Then I will sketch the graph carefully on the interval $[1,2]$, labeling at least the points $(1,1)$ and $(2,\frac{1}{2})$. Then I will ask them if they can see some reason to believe that the area under h between 1 and 2 is more than $\frac{1}{2}$. If I do not get a prompt response I may ask if they can see why the area must be less than 1. I have always had someone go to the board and draw an appropriate rectangle. If I did not, then I would ask if someone would go to the board and draw the simplest lower Riemann sum they can think of for h on the interval $[1,2]$. Once we have the two rectangles enclosed by and enclosing the graph of areas 1 and $\frac{1}{2}$, if this has not yet been done, I will point out that they are the simplest possible Riemann sums and the partition of the interval $[1,2]$ is the two term sequence $1,2$.

Next I will ask them to find the best possible lower Riemann sum for $\int_1^2 h$ using only 3 partition points. Timing here is critical. I want to be sure I ask this question before the class is over, even if that means skimping on some of the details described above.

Their response will determine what I do next. If they indicate that they have no idea how to do the problem I am likely to act surprised and remind them that they know how to do max-min problems and that this is just finding the maximum value of all the lower Riemann sums with 3 partition points. If necessary I will label the partition $1,x,2$ where $1 < x < 2$. If someone says that it is surely in the middle at 1.5, then I will likely suggest they get out their calculator, and I will get mine out and compute the values for $x = 1.4$, $x = 1.5$ and $x = 1.6$.

Depending on how much time we have I will ask now or later if someone can see why $\int_1^2 h < \frac{3}{4}$, pointing out that this is the average of the two upper and lower Riemann sums.

Or I may ask them to work the same problem for f instead of h where $f(x) = \frac{1}{x^2}$ or $f(x) = x$ or $f(x) = x^2$ for $1 \le x \le 2$.

At the end of the first period I will hand out the following notes.

You are probably used to writing the integral from a to b of the function h as $\int_a^b h(x)dx$. We see no need for the x or the dx and will write this integral as $\int_a^b h$.

You surely think of $\int_a^b h$ as the area between the graph of h and the $x-axis$ between the vertical lines through the points $(a,0)$ and $(b,0)$. We will also think of the integral as being this area.

We study Riemann sums because they give us a way of approximating an integral. Indeed, we will define an integral in terms of Riemann sums. Geometrically a Riemann sum is just the sum of the areas of a collection of non-overlapping rectangles. In the case of h to get an upper Riemann sum for $\int_1^2 h$ we divide the interval $[1,2]$ into non-overlapping intervals and for each of these intervals we form a rectangle whose base is that interval and whose height is the biggest value of $h(x)$ for x in that interval. Since h is a decreasing graph, this would be at the left endpoint of the interval. The upper Riemann sum is the total area enclosed by these rectangles. A lower Riemann sum is gotten in the same way except we use the smallest value of $h(x)$ for x in a given interval for the height. For h this would be at the left endpoint. Later we will define these sums analytically as you probably did in your AP calculus. Geometrically it should be clear that each lower Riemann sum for h on $[a,b]$ is less than $\int_a^b h$ and each upper Riemann sum is larger than $\int_a^b h$.

The simplest type of Riemann sums are those with only two partition points so that they determine only one rectangle. Explain why the values of the upper and lower Riemann sums for $\int_1^2 h$, on a partition with only 2 points are 1 and $1/2$. Later we will see that this integral gives us the natural logarithm function. At this point we do not have a definition for $\int_1^2 h$, but thinking of the integral as an area explain why $\frac{1}{2} < \int_1^2 h < 1$.

Explain why $\int_1^2 h < \frac{3}{4}$. Do you see that this is because the graph of h is concave up?

Problem 1. *Find a partition $P = x_0, x_1, x_2$ of $[1,2]$ such that the lower Riemann sum for $\int_1^2 h$ on P is larger than any other lower Riemann sum for $\int_1^2 h$ on a partition with only 3 partition points.*

- This problem will surely be done by someone at the 2nd class meeting. It and the following similar problems give an opportunity for the instructor to learn something about the students and for the student to learn something. But one must be careful not to put a student down. Once the student has done the work to determine a local max or min, by finding a formula for the lower Riemann sum $A(x)$ for $0 < x < 1$ they can be asked how they know the place where $A'(x) = 0$ yields a maximum. This could and hopefully will start some discussion about what they learned in high school and it will probably be good to discuss why their rules apply. I do not ask them to show that any rules they quote are correct but I discuss any rules they state that are not correct. At this point I accept the intermediate value theorem or that a function with positive derivative on an interval is increasing on that interval or that a maximum cannot occur at a point where the derivative is positive or negative (except at an endpoint). What I am most interested in at this point is to see if they can give a correct reason for believing they have a maximum.

Problem 2. *Find a partition $P = x_0, x_1, x_2$ of $[1,2]$ such that the upper Riemann sum for $\int_1^2 h$ on P is smaller than any other upper Riemann sum for $\int_1^2 h$ on a partition with only 3 partition points.*

Note that these Riemann sums give the best upper and lower Riemann approximations for $\int_1^2 h$ with only 3 partition points.

The next problem has two parts which are very much alike. In general for problems like this one I will have different students present each part. And if you decide to write up one of these problems, you do not have to do both parts.

Problem 3. *Let f be the function such that $f(x) = x$ for each number x such that $1 \le x \le 2$. Find the best upper and lower approximations for $\int_1^2 f$ with only 3 partition points.*

Problem 4. *Let f be the function such that $f(x) = x^2$ for each number x such that $1 \le x \le 2$. Find the best upper and lower approximations for $\int_1^2 f$ with only 3 partition points.*

Chapter 2

The natural logarithm function

Definition 2. *If $[a,b]$ is an interval then P is called a* **partition** *of the interval $[a,b]$ if there is a positive integer n such that P is a finite increasing sequence of n numbers with a the first number and b the last.*

Notation 2. *If n is any positive integer, we will usually denote a* **partition** *P having $n+1$ points by $\mathbf{x_0}, \mathbf{x_1}, \mathbf{x_2}, ..., \mathbf{x_n}$ where $x_0 = a$, $x_n = b$, and if j is an integer from 1 to n, then $x_{j-1} < x_j$. We often think of P as a way of dividing the interval $[a,b]$ into the n non-overlapping intervals $[x_1,x_2],[x_2,x_3],\cdots,[x_{n-1},x_n]$.*

Review Previously we noted that the upper and lower Riemann sums for $\int_1^2 h$ with only two partition points were $\frac{1}{2}$ and 1 so that we had $.5 < \int_1^2 h < 1$. This was because the upper Riemann sum came from one rectangle which enclosed the area represented by $\int_1^2 h$ and the lower Riemann sum came from one rectangle which was enclosed by this area.

Then we noted that there was a trapezoid of area $\frac{3}{4}$ which enclosed this area. You probably had a formula for the trapezoid rule which would give this value. So now we have that $.5 < \int_1^2 h < .75$.

The word calculus comes from the same stem as calculate and that is what we will be doing most of this semester. Assuming for these calculations that $\ln 2 = \int_1^2 h$, then we could average our two approximations .75 and .5 to get .625 as an approximation for $\ln 2$ with an accuracy of $\pm$ .125. Note that it is of little use to say that $\ln 2$ is approximately .625 unless you specify the accuracy of the approximation. It is sometimes very hard to determine the accuracy of a given approximation. Your calculator will indicate $\ln 2 = .69$ to two decimal places so this fits with our data.

Here is our definition of a Riemann sum. As you read this definition, let $n = 3$, draw a graph of h and represent the lower Riemann sum for $\int_1^2 h$ as the sum of the areas of 3 rectangles.

Definition 3. *A* **lower Riemann sum** *for $\int_a^b h$ is a number obtained as follows. First select an integer n and a partition $P = x_0, x_1, x_2, \cdots, x_n$ of an interval $[a,b]$ consisting of $n+1$ points. Then for each positive integer i, $0 \le i < n$, form a rectangle whose base is the interval $[x_i, x_{i+1}]$ and whose height is the smallest value of the function h in the interval $[x_i, x_{i+1}]$. Then form the sum of all the rectangles. That number is a lower Riemann sum. To get an upper Riemann sum do exactly the same except that the height of each rectangle* *should be the largest value of the function h in the appropriate interval. For the function h this would be at the left endpoint so you could use your formulas from your AP calculus to check that your answers are correct. But you should be able to explain at the board how to get these formulas from the sum of the rectangles described above.*

Problem 5. *Find the best lower Riemann approximation for $\int_1^2 h$ with 4 partition points and 3 rectangles.*

- The preceding problem can be fun. I once had a superior student instantly say: "We can't do that problem without multivariable calculus!". I asked if he had taken multivariable calculus and when he said no I jokingly asked how he knew that he needed it. I would not have made such a sharp remark to a weaker student. And hints are generally needed on this problem. One must make a judgment call here. If I think they need help I might suggest that they pick a fixed partition point, say $\frac{3}{2}$, and then ask what is the best choice for another one between $\frac{3}{2}$ and 2. Or just call the first one a with $1 < a < 2$ and the next one $x \in (a, 2)$. and find the best choice for x in terms of a. This is just an example of what I might do, depending of what is said in class. But one must work hard at convincing the students that they know enough to solve the problems I am giving them. Another, more abstract hint would be to ask if they can find the best lower Riemann approximation for $\int_a^b h$ with 3 partition points, in terms of a and b of course, and then use that to work the preceding problem.

  Also, depending on the syllabus and time this is an interesting problem to pursue. In one class I had a student observe that the areas of the rectangles were the same in each of the two cases we had considered, something I did not know. We went on to show that this is true for each choice of n which gave a very simple sequence converging to $\ln 2$. And it is a good place for students to spot a pattern and guess a theorem. With 2 rectangles the partition is $1, \sqrt{2}, 2$ and with 3 it is $1, 2^{\frac{1}{3}}, 2^{\frac{2}{3}}, 2$. It is not hard to get them to guess the answer for 4 and prove it is correct, but it does take class time from "covering material". This is done in several later problems which can, of course, be omitted at the instructor's discretion.

For many of the following problems, you are asked to do something for an integer n, and you may not see how to do this. In that case, work the problem for $n = 2$, then $n = 3$ and look for a pattern and see if you can guess an answer for $n = 4$ or for any choice of n. We can discuss in class how to do it for any choice of n.

- This is another case where one can spend a lot of time or a little, depending on what must be covered. Once the students see a pattern they will have no doubt of the final formula. But unless one wants to teach induction they may have no way of showing their formulas are correct. I get them to describe what they did for $n = 2$ or $n = 3$. Then, for the following problem as an example, for an arbitrary choice of n with $n+1$ partition points, I may have them label the partition points, and the width of each rectangle in terms of n and then write out the lower Riemann sum either using summation notation or by writing out the first few terms, followed by $\cdots$ followed by the last term or two.

Problem 6. *Let n be a positive integer. Find a formula for the lower Riemann sum for $\int_1^2 h$ using n rectangles of equal width and $n+1$ partition points. If you do not know how to get* *a formula that works for any choice of n, then do it for $n = 2$, then $n = 3$ and see if you can guess a formula for $n = 4$. Then check it for $n = 4$, and if it is correct then try to guess one that works for any choice of n.*

Problem 7. *Do the same as in Problem 6 except do it for the upper Riemann sums.*

Problem 8. *Let n be an arbitrary positive integer. Find a formula for the difference between the upper and lower Riemann sums for $\int_1^2 h$ with $n+1$ equally spaced partition points. Note that this will tell us how good (or bad) our approximation might be.*

- I sometimes will ask if they can use the result of the preceding problem to tell me how many rectangles are needed to get an approximation for $\ln 2$ that is accurate to 3 decimal places.

Problem 9. *Let n be any positive integer. Show that the lower Riemann sum for $\int_1^2 h$ on a partition with n equally spaced partition points is smaller than the lower Riemann sum for $\int_1^2 h$ on a partition with $n+1$ equally spaced partition points.*

- The next problem is not needed in the course but came up when one student asked if we could use a trapezoid type approximation for a lower approximation to an integral to get a better approximation. It is an interesting problem and gives an easy way to show that $\ln 3 > 1$, so $e < 3$.

Problem 10. *Find the best trapezoidal approximation for $\int_1^2 h$ where the trapezoid lies on or under the graph of h, its base is the interval $[1,2]$ and the top is tangent to the graph h.*

Problem 11. *Let a and b be numbers with $0 < a < b$. Find a formula for the lower Riemann sum for $\int_a^b h$ using n equal width rectangles.*

Problem 12. *Let a and b be numbers with $0 < a < b$, and let c be a positive number. Show that the lower Riemann sum for $\int_{ca}^{cb} h$ using n equal width rectangles is the same as that for $\int_a^b h$ using the same number of rectangles from the previous problem.*

If you don't see how to do the preceding problem, then find the lower Riemann approximation for $\int_2^4 h$ using n equal width rectangles and show that it is the same as that from Problem 6. Then do the same for $\int_3^6 h$ and see if this helps on the original problem.

Problem 13. *Let a and b be numbers with $0 < a < b$ and let c be a positive number. Show that the upper Riemann approximation for $\int_a^b h$ with n equal width rectangles is the same as the upper Riemann approximation for $\int_{ca}^{cb} h$ with n equal width rectangles.*

The following definition should have been given earlier when we started talking about Riemann approximations, but we did not need it. Now we do.

Definition 4. *If a and b are two numbers with $0 < a < b$ then the integral from a to b of h, written $\int_a^b h$ is the only number which is larger than each lower Riemann approximation for $\int_a^b h$ with equally spaced partition points and also smaller than each upper Riemann approximation for $\int_a^b h$ with equally spaced partition points.*

I hope that you would agree that if $\int_a^b h$ is to represent the area between h and the x-axis between a and b, then it should be larger than each of our lower Riemann approximation and smaller than each of the upper Riemann approximations. We could actually prove that there is only one such number but I don't see much point in doing that at this time. I am more interested in showing how to use these approximations to learn something about the integral.

Now for the problem I have been aiming for. There is really not anything to prove in Problem 14 as it simply follows from the Definition 4 and the preceding 3 problems.

Problem 14. *If a, b, and c are positive numbers and $a < b$, then $\int_a^b h = \int_{ca}^{cb} h$.*

Now we want to take care of the case where $a = b$ and $a > b$.

Definition 5. *Assume that each of a and b is a positive number and $a > b$. Define $\int_a^a h = 0$, and $\int_a^b h = -\int_b^a h$,*

We will need one more property of integrals which you surely were told about in high school and we could prove but we will just accept and use it. I hope it is clear that this should be true and I will not ask you to do the following problem. We call it a theorem because it is true and we could show it, but it would take too much time. So we will just accept it and use it as needed.

Theorem 1. *If a, b and c are numbers and $a < b < c$, then $\int_a^b h + \int_b^c h = \int_a^c h$.*

In case $a < b < c$ it should be clear that Theorem 1 should be true if we think of the integrals as areas. But it works regardless of how a, b and c are related. You should be able to use Problem 1 and Definition 5 to do the next problem.

Problem 15. *If a, b and c are positive numbers and $a < c < b$, show that $\int_a^b h + \int_b^c h = \int_a^c h$.*

There are lots of other cases to consider but they can all be done in much the same way as the preceding problem. So we will just accept and use the fact that Problem 1 works regardless of what number is represented by each of a, b and c. They do not even have to be different numbers. The following problem states that fact for reference purposes. I will not ask anyone to present this one.

Problem 16. *If each of a, b and c is a positive number then $\int_a^b h + \int_b^c h = \int_a^c h$.*

- The next 5 problems are here because some of my past students discovered them. They can be included or not. They have no bearing on the rest of the course. I discussed this earlier and these problems could have been given earlier but I felt it better to put it off until the students had more practice looking for patterns.

Problem 17. *Show that the 2 rectangles in Problem 1 have the same area and that the 3 rectangles from Problem 5 have the same area.*

We have guessed in class that the best lower Riemann approximation for $\int_1^2 h$ with 4 rectangles is given by using the partition $(1, 2^{\frac{1}{4}}, 4^{\frac{1}{4}}, 8^{\frac{1}{4}}, 2)$. So if n is a positive integer and $n > 1$, then we would probably guess that the best lower Riemann approximation with n rectangles would be given by using the partition $(1, 2^{\frac{1}{n}}, 2^{\frac{2}{n}}, 2^{\frac{3}{n}}, \cdots, 2^{\frac{n-1}{n}}, 2)$.

Problem 18. *Show that if n is a positive integer, then in the lower Riemann approximation for* $\int_1^2 h$ *using the partition* $(1, 2^{\frac{1}{n}}, 2^{\frac{2}{n}}, 2^{\frac{3}{n}}, \cdots, 2^{\frac{n-1}{n}}, 2)$, *the rectangles all have the same size.*

Problem 19. *Show that if n is a positive integer then in the upper Riemann approximation for* $\int_1^2 h$ *using the partition* $(1, 2^{\frac{1}{n}}, 2^{\frac{2}{n}}, 2^{\frac{3}{n}}, \cdots, 2^{\frac{n-1}{n}}, 2)$, *the rectangles all have the same size.*

Problem 20. *Use the results of the preceding problem to show that if n is a positive integer and* $n > 1$ *then the average of the upper and lower approximations for* $\int_1^2 h$ *using the partition* $(1, 2^{\frac{1}{n}}, 2^{\frac{2}{n}}, 2^{\frac{3}{n}}, \cdots, 2^{\frac{n-1}{n}}, 2)$ *is* $\frac{n}{2}(2^{\frac{1}{n}} - 2^{-\frac{1}{n}})$.

Problem 21. *This is not one to be written up but use your calculator to compute the approximation for* $\int_1^2 h$ *given in the preceding problem for some large value of n. Say* 1000 *or* 10000 *and compare your answer with* $\ln 2$ *from your calculator. Your answer should be too large. Do you see why?*

Notation 3. *We are now ready to define the natural logarithm function as* $\ln x = \int_1^x h$ *for* $x > 0$. *We prefer to have a single letter to represent this function so we will define* $L(x) = \ln x$.

As a reminder, for the following problems about the function L, you are not to use any properties of $\ln x$ from your high school calculus. Use only Definitions 4, 5 and 3, and Problems 14 and 16.

- This is another case where the instructor must decide how many problems like the following will be stated. It depends on the class. I have found that they enjoy working on these problems and some can be quite challenging. But this is another case of working out special cases and then moving to the main theorems.

Problem 22. *Show that* $L(2) + L(3) = L(6)$. *That is, show that* $\int_1^2 h + \int_1^3 h = \int_1^6 h$.

Problem 23. *Show that* $\ln 4 + \ln 5 = \ln 20$.

Problem 24. *Show that* $\ln \frac{1}{3} = -\ln 3$

Problem 25. *Show that* $\ln 2 = \frac{1}{2}\ln 4$

- This is a particularly challenging problem.

Problem 26. *Show that* $\ln(\sqrt{2})^3 = \frac{3}{2}\ln 2$

In the following problems, if you don't see how to do them, then substitute values for a or b and see if you can work them then. If you can, see if the same method works for any choice of a or b.

Problem 27. *If a is any positive number and b is any positive number then* $\ln a + \ln b = \ln ab$.

Problem 28. *Show that if a is any positive number and b is any positive number, then* $\ln \frac{a}{b} = \ln a - \ln b$.

- I sometimes include the following problem to see if anyone in the class will recognize that they could just use the preceding problem with $a = 1$.

Problem 29. *Show that if a is any positive number then* $\ln \frac{1}{a} = -\ln a$.

For the problems which involve an integer n, do them for $n = 1$ and $n = 2$ and $n = 3$. Then decide if your method would work for any choice of n.

- As was the case before, without induction the students can't really show that these formulas work or any choice of the integer n. I am satisfied with letting them describe a method which we all agree would work for any choice of n.

Problem 30. *Show that if a is any positive number and n is any positive integer, then* $\ln a^n = n \ln a$.

Definition 6. *The statement that a is a* **rational** *number means that there is an integer* $n \neq 0$ *and an integer m such that* $a = \frac{m}{n}$.

You have been using some of the following facts about numbers. I list them here just as a review and to remind you that you do not know that they are valid unless the exponents are rational numbers. You surely have never seen a definition for 2^x in case x is an irrational number, for example: $2^{\sqrt{2}}$.

For the following problems you can use basic algebraic facts about exponents when the exponent is a **rational** number. So if x is a number, $x \neq 0$, and a and b are rational numbers, then
1) $x^a x^b = x^{a+b}$, and $x^a/x^b = x^{a-b}$,
2) $x^{-a} = \frac{1}{x^a}$, and
3) $(x^a)^b = x^{ab}$

In fact some of these rules are valid even if $x = 0$ depending on what a and b are.

Problem 31. *Show that if a is any positive number and n is any positive integer, then* $\ln a^{\frac{1}{n}} = \frac{1}{n} \ln a$.

Problem 32. *Show that if a is any positive number and n is any positive integer and m is any positive integer, then* $\ln a^{\frac{m}{n}} = \frac{m}{n} \ln a$.

Problem 33. *Show that if a is a positive number and r is any rational number, positive, negative or zero, then* $\ln a^r = r \ln a$.

Note that we can't show that if a is a positive number then $\ln a^{\sqrt{2}} = \sqrt{2} \ln a$ because we do not have a definition for a^x when x is an irrational number. First we need a definition for an irrational number. That much is easy.

Definition 7. *The statement that the number x is irrational means that x is not a rational number.*

You probably do not know that any number is irrational so let's work on the following problem.

- The following problem is one the students seem to really enjoy working on but again it is not of any importance to our problem sequence. I have seldom had a student handle the case where both were odd without more hints than are given in the notes.

Problem 34. *Show that* $\sqrt{2}$ *is an irrational number.*

If you can do the preceding problem, present it in class. If you want help, read on.

Again we need some definitions before proceeding.

Definition 8. *The statement that n is an* **even** *integer means that there is an integer k such that* $n = 2k$*. The statement that n is an odd integer means that there is an integer k such that* $n = 2k+1$.

To work Problem 34 you must show that there do not exist two integers m and n such that $\frac{m}{n} = \sqrt{2}$. And you can assume that m and n are not both even integers, since, if they were then you would have $m = 2m'$ and $n = 2n'$ for some integers m' and n' and also $\frac{m'}{n'} = \sqrt{2}$. And if m' and n' were both even you could repeat this process continuing until you get integers whose quotient is $\sqrt{2}$ which are not both even.

So, to get started on Problem 34 try the following one as a start.

Problem 35. *Show that if m is an odd integer and n is an even integer, then* $\frac{m^2}{n^2} \neq 2$.

- It is in trying to get them to use this definition in the next problem that I first discovered how little they understand about inequalities or how to work with them. I strongly feel that some axioms to be used and some problems involving inequalities are badly needed in the course.

Definition 9. *The statement that* f *is an increasing function means that if a and b are in the domain of* f *and* $a < b$ *then* $f(a) < f(b)$.

Problem 36. *Show that L is an increasing function. To show this, you must use our definition for* $L(x)$ *for* $x > 0$*. Use a Riemann sum to show that if a and b are positive numbers, and* $a < b$*, then* $L(b) - L(a) > 0$.

We now know that L is an increasing function and we know that the domain of L is the set of all numbers. Does this mean that L crosses every horizontal line?

- I have been surprised at how bad these students' intuition is about the behavior of a graph f as $x \to \infty$. They are likely to claim that $f(x)$ must approach ∞ if f is an increasing graph. And that $f(x)$ does not approach ∞ if $f'(x) \to 0$ as $x \to \infty$.

Problem 37. *Find a graph* f *whose domain is the set of all numbers which is an increasing graph and which lies entirely below the horizontal line with equation* $y = 1$.

This completes most of our study of the function L. Next we want to investigate properties of the inverse of L.

Chapter 3

The exponential function

It seems to me to be surprising that the range of the graph L is the set of all numbers. Use your graphing calculator to graph $L(x) = \ln x$ and see if you think the graph will cross the line with equation $y = 10$. You surely recall from your calculus that $L'(x) = 1/x$ for each positive number x. From this it follows that $L'(x)$ approaches zero as x increases without bound.

Problem 38. *Show that* $\ln 4 > 1$.

You probably had the following Theorem in high school. It is called the intermediate value theorem and is one of those things that is "obvious" but not easy to prove. We will just assume that it is true without proof. We only need it right now for the function L but it is valid for any continuous function.

Theorem 2. *If each of a, b and y is a number, $a < b$ and $L(a) < y < L(b)$, then there is a number x with $a < x < b$ such that $L(x) = y$.*

We are now ready to define the base of our logarithm function L.

Problem 39. *Recall that we showed earlier that $L(2) < 1$. Explain how you know from Theorem 2 and Problem 38 that there is a number which we will call e such that $L(e) = 1$.*

Problem 40. *Show that if n is a positive integer, there is a number x such that $L(x) = n$. As usual, if you don't see how to do this, try it for $n = 2$ or $n = 3$ and see if that helps.*

For the next problem we will need the following property about numbers which I hope you will agree is "obvious". We call it an Axiom to indicate that we think it would be difficult to prove it on the basis of anything simpler, so we just assume it to be true without proof.

Axiom 1. *If x is a number, then x is an integer (positive, negative or 0) or else there is an integer n such that $n < x < n+1$.*

Problem 41. *Show that if y is a positive number, then there is a number x such that $L(x) = \ln x = y$. Hint: Use Theorem 2 and Axiom 1.*

Definition 10. *Recall that a* **function** *is simply a collection of ordered pairs (x,y) of numbers such that no two of them have the same first term. Or, if you prefer, a function is a* *collection of points in a plane no two of which lie on the same vertical line. If f is a function then by the* **domain** *of f is meant the set of all numbers which are the first terms (the x-coordinates) of points on f. And the* **range** *is the set of all 2nd terms (y-coordinates) of points of f.*

Definition 11. *If f is an increasing or a decreasing function, then by the* **inverse** *of f is meant the set of all points (b,a) such that (a,b) is a point of f.*

Problem 42. *Show that if f is an increasing or decreasing function, then the inverse of f is a function.*

Problem 43. *If f is an increasing or decreasing function and g is the inverse of f, use the definition and the usual notation of $(x, f(x))$ being a point on f, explain why $g(f(x)) = x$ if x is in the domain of f and $f(g(x)) = x$ if x is in the domain of g.*

Definition 12. *E will denote the inverse of L.*

You know about exponents when the exponent is a rational number. So you know what is meant by e^3 or $\sqrt{e}$ or $e^{\frac{2}{3}}$. And you know what $E(x)$ means for any number x, rational or irrational, from the definition of E as the inverse of L. Our next set of problems will be to show that if r is a rational number then $E(r) = e^r$. So at least for rational numbers the graph of E is the same as the graph of the function $y = e^x$. We will begin as we did with the natural logarithm function by showing this for integers.

Using what we know about L it is not hard to show what we want about its inverse E since from the definition of inverse, if (x,y) is on L or $L(x) = y$, then (y,x) is on E so $E(y) = x$.

Problem 44. *Explain why $E(0) = e^0$ and $E(1) = e^1$ just from the definition of E and e and what you know about exponents.*

Problem 45. *Show that if n is a positive integer $E(n) = e^n$.*

Problem 46. *Show that if n and m are positive integers, then $E(\frac{m}{n}) = e^{\frac{m}{n}}$.*

Problem 47. *Show that if r is a rational number then $E(r) = e^r$.*

Now that we know that $E(r) = e^r$ if r is a rational number and we know that the usual rules for exponents work for rational numbers then we know that if each of a and b is a rational number then we can express these rules in terms of E. So $E(a) \cdot E(b) = E(a+b)$ $E(-a) = 1/E(a)$, and $E(a)^b = E(ab)$.

Actually it is easy to show that these rules work for any choice of a and b, using what we know about the graph $L(x) = \ln x$. Here are a couple of examples.

Problem 48. *Show that if each of a and b is a number, then $E(a+b) = E(a) \cdot E(b)$.*

Problem 49. *Show that if a is a number, then $E(-a) = 1/E(a)$.*

We are finally ready to define what we mean by e^x if x is an irrational number. We **define** e^x to mean $E(x)$ which is the number y such that $L(y) = x$. Noting that if $a > 0$, then $e^{\ln a} = E(L(a)) = a$ and we want $a^b = (e^{\ln a})^b = e^{b \ln a}$, we make the following definition.

Definition 13. *If $a > 0$ and b is a number, $a^b = E(b \ln a)$.*

- I tried to do all of this carefully for a couple of semesters and decided that it was entirely too subtle for most of these students. As a result it took entirely too much time that I feel could be better spent on other topics. So I decided to omit some of the details surrounding rational and irrational exponents. The students seem happy to just believe that all the rules work for irrational numbers.

Now we have the problem of showing that this definition agrees with our usual one. This is so much like what we have already done that we will just assume that the rules you are familiar with work not only for rational numbers. but irrational ones as well.

For the following problems I will assume that you know the formulas for the derivatives of polynomials and trigonometric functions. I don't mean you have to memorize them. You can look them up as needed. We will also use the various rules for computing derivatives such as the product, sum, quotient and chain rules. The chain rule is often written in different ways. Here is the form we will use. If f and g are functions and the range of g is the domain of f and $h(x) = f(g(x))$ for each number x in the domain of g, then under appropriate conditions, if x is in the domain of h, $h'(x) = f'(g(x)) \cdot g'(x)$. We will just assume this works for all the functions we will consider.

You surely recall that $L'(x) = 1/x$ and $E'(x) = E(x)$. We could use Riemann sums again to show that if $x > 0$, $L'(x) = 1/x$. Instead we will use one of the fundamental theorems of calculus. The following is one of them.

Theorem 3. *If f is a continuous function whose domain includes the interval $[a,b]$, and c is a number in $[a,b]$, and $A(x) = \int_c^x f$ for each number x in $[a,b]$, then for each number x in $[a,b]$, $A'(x) = f(x)$.*

It follows from Theorem 3 that $L'(x) = h(x) = 1/x$ for each number $x > 0$. We will use this fact from now on.

Chapter 4

Taylor Polynomials

Problem 50. *Use the definition of E, the chain rule and the fact that $L'(x) = \frac{1}{x}$ for each number $x > 0$ to show that $E'(x) = E(x)$ for each number x.*

Problem 51. *Find the tangent line to the graph of E at the point $(0,1)$.*

Problem 52. *Find a "tangent parabola" to the graph of E at the point $(0,1)$. That is, find a parabola P such that P and E have the same value, derivative and 2nd derivative at the point $(0,1)$.*

Definition 14. *If n is a positive integer, the statement that P is a* **polynomial of degree n** *means that there are numbers $a_0, a_1, \ldots, a_n$ such that $a_n \neq 0$ and for each number x, $P(x) = a_0 + a_1x + a_2x^2 + \ldots a_nx^n$. Or, if you prefer, $P(x) = \sum_{k=0}^{n} a_kx^k$.*

Problem 53. *Find a polynomial P of degree 3 such that $P(0) = E(0)$, $P'(0) = E'(0)$, $P''(0) = E''(0)$ and $P'''(0) = E'''(0)$. Can you find a formula for a polynomial P of degree n so that $P(0) = E(0)$ and the first n derivatives of P and E are the same at the point $(0,1)$?*

The polynomials in Problem 53 are called the **Taylor polynomials** for E of degree 3 and n based at 0.

- I do not make an issue out of the uniqueness of the Taylor polynomials, but I sometimes point out that there is only one tangent line at a given point after they find it. And I may do the same for the tangent parabola. They often ask a bunch of questions about whether one has the same polynomial or not when they write the polynomial in powers of x or powers of $x-1$. This is usually a good discussion from which they learn a lot.

Next we want to find Taylor polynomials for E based at 1.

Problem 54. *Find the tangent line to E at the point $(1,e)$. Find a and b so that your tangent line has equation $y = a + b(x-1)$.*

Problem 55. *As in Problem 52 find a tangent parabola to the graph of E at the point $(1,e)$. That is, find a polynomial P of degree 2 such that $P(1) = E(0)$, $P'(1) = E'(1)$ and $P''(1) = E''(1)$. As in the preceding problem, write your answer in the form $a + b(x-1) + c(x-1)^2$. This is the Taylor polynomial of degree 2 based at 1 for E.*

Notation 4. *If n is a positive integer and f is a function and x is a number in the domain of f at which f has n derivatives, then $f^{(n)}(x)$ denotes the n^{th} derivative of f at x.*

Question 1. *Can you give an example of a function f which you think is continuous at some number x in its domain but does not have a derivative there? No proof required, nor could you give one without a definition for continuous and for derivative. Can you describe a function f that has a derivative at some number x but does not have a 2nd derivative at x?*

Definition 15. *If f is a function and n is a positive integer and a is a number in the domain of f and f has a jth derivative for each positive integer $j \leq n$, then the* **Taylor polynomial** *for f based at a is the polynomial P of degree n such that if $0 \leq j \leq n$, then $f^{(j)}(a) = P^{(j)}(a)$.*

Problem 56. *Let n be a positive integer. Find numbers $a_0, a_1, a_2, \cdots, a_n$ such that if $P_n(x) = a_0 + a_1(x-1) + a_2(x-1)^2 + \cdots a_n(x-1)^n$ for each number x, then P_n is the Taylor polynomial of degree n based at 1 for E.*

Problem 57. *Show that there is no Taylor polynomial of degree 2 for f based at 0 if $f(x) = \sin x$ for each number x.*

Problem 58. *Let n be an odd positive integer. Find the Taylor polynomial of degree n based at 0 for f if $f(x) = \sin x$ for each number x. I don't expect you to prove that your formula is correct. Work the problem for $n = 3$ and $n = 5$ and see if you can find a pattern.*

Problem 59. *Let $f(x) = \cos x$ for each number x. Let n be an even positive integer. Find the Taylor polynomial of degree n for f based at 0. Again. as in the previous problem, find a pattern.*

Question 2. *If n is a positive integer, are you ready to guess a formula which gives the Taylor polynomial of degree n based at the number a for any function f for which you can find the derivatives $f^j(a)$ for values of j from 1 to n?*

- I am not sure what to do about the next one. They are not likely to get the Taylor polynomial in "standard" form in powers of $x - \pi/2$ because they find it more natural to compute, for small values of n, the coefficients in powers of x. Perhaps one should just lecture briefly about the standard form. I changed several statements in the preceding problems hoping to solve this problem. They really should see the formula for the Taylor series in standard form.

Problem 60. *Find the Taylor polynomial of degree 2 for f based at $\frac{\pi}{2}$ if $f(x) = \cos x$ for each number x.*

Problem 61. *Find the Taylor polynomial of degree 4 for f based at $\frac{\pi}{2}$ if $f(x) = \cos x$ for each number x. Use it and the polynomial of degree 4 from Problem 59 to compute an approximation for $\cos 1.5$. Which is the best approximation and why do you think it should be?*

Problem 62. *Let $f(x) = \tan x$ for each number x. Find the Taylor polynomial of degree 3 for f based at 0. I don't know of any pattern.*

Chapter 5

Series and sequences

By an **increasing sequence** we mean an unending sequence of numbers $x_1, x_2, \ldots$ such that for each positive integer n, $x_n < x_{n+1}$. A **decreasing sequence** is defined in a similar way. An increasing sequence is said to be **bounded** if there is a number larger than each number in the sequence. As an example, the sequence $x_1, x_2, x_3, \ldots = 1, 2, 3, \ldots$ is not bounded since for each positive integer n, $x_n = n$. and by Axiom 1 for any number x, there is a positive integer $n+1$ which is larger than x. On the other hand the sequence $x_1, x_2, x_3, \ldots = 1/2, 2/3, 3/4, \ldots$ is bounded since for each positive integer n, $x_n = 1 - \frac{1}{n+1} < 2$. And it is increasing by the next problem.

- This might be a good place to insert some problems about inequalities, using some axioms. I may try the following axioms next time. If so I will add some problems about inequalities.

Here are all the rules you need to be able to work with inequalities:

Assume that each of a, b and c is a number.

1) If $a \neq b$, then $a < b$ or $b < a$.
2) If $a < b$ and $b < c$, then $a < c$.
3) If $a < b$ then $a + c < b + c$.
4) If $c > 0$ and $a < b$, then $ca < cb$.
5) If $c < 0$ and $a < b$, then $ca > cb$.
6) $0 < 1$.

I hope these are all familiar. You should be able to get other familiar facts from these. Use them on the next problem.

Problem 63. *Use the preceding rules about inequalities to show that if n is a positive integer,* $1 - \frac{1}{n+1} < 1 - \frac{1}{n+2}$.

Consider the following sequence: $1/2, 3/4, 7/8, 15/16, \ldots$. The number 2 is larger than every number in this sequence. But 2 is not the smallest such number. The number 1 is also larger than each number in the sequence. We will later see how to show that 1 is the smallest such number.

We now need another Axiom which I hope seems reasonable to you.

Axiom 2. *If $x_1, x_2, \ldots$ is an increasing (or decreasing) sequence of numbers and there is a number which is larger (smaller) than every number in the sequence, then there is a smallest (largest) such number. It is called the* **limit** *of the sequence.*

Problem 64. *Show that 0 is the limit of the sequence $1, 1/2, 1/3, \ldots$. Note that clearly 0 is smaller than each number in the sequence so we must show that no positive number is smaller than each number in the sequence. In other words. we want to know that if a is any positive number then a is not smaller than each number in the sequence. This means that you must show that if a is any positive number, then there is an integer N such that $1/N$ (which is a term in the sequence) is smaller than or equal to a. Hint: Use Axiom 1.*

Definition 16. *A sequence is said to be* **bounded above** *if there is a number which is larger than every number in the sequence, and* **bounded below** *if there is a number smaller than every number in the sequence.*

We often describe a sequence by starting with a number and then adding numbers over and over. For example the sequence: $1/2, 3/4, 7/8, 15/16, \ldots$ could be written as $1/2, 1/2 + 1/4, 1/2 + 1/4 + 1/8, \ldots$. We often abbreviate this sequence as $1/2 + 1/4 + 1/8 + 1/16 + \ldots$. When written like this we call it a **series** but it is just a different way of describing a sequence. Books would refer to the sequence as the sequence of partial sums: $1/2, 1/2 + 1/4, 1/2 + 1/4 + 1/8, \ldots$. By the limit of a series we mean the limit of the corresponding sequence.

Definition 17. *The statement that a series or a sequence* **converges** *means that it has a limit.*

So the problem of deciding if an increasing sequence converges is a matter of finding out if it is bounded above. This leads us to what is called the **integral test**.

For the rest of this course we will frequently use the fundamental theorems of calculus. One of them, Theorem 3 was stated and used earlier to show that $L'(x) = \frac{1}{x}$ for $x > 0$. The other one is stated next. We will not prove this.

Theorem 4. *If a and b are numbers with $a < b$ and f is a function and g is a continuous function such that $g(x) = f'(x)$ for each number x such that $a \leq x \leq b$, then $\int_a^b g = f(b) - f(a)$.*

We will assume that all the functions we consider are continuous so we can use Theorem 4. You can also use what you know about derivatives and anti-derivatives or indefinite integrals for any of the rest of our problems.

Problem 65. *Let $f(x) = \frac{1}{x^2}$ for each number $x > 0$. Use Riemann sums to show that $1/2^2 < \int_1^2 f$ and $\frac{1}{2^2} + \frac{1}{3^2} < \int_1^3 f$. Let n be a positive integer. Show that $1/2^2 + 1/3^2 + 1/4^2 + \ldots + 1/n^2$ is a lower Riemann sum for $\int_1^n f$.*

Problem 66. *Show that if f is the function so that $f(x) = \frac{1}{x^2}$ for each number $x > 0$, and n is any positive integer, then $\int_1^n f < 1$.*

From the preceding two problems we have that the sequence $1/2^2 + 1/3^2 + 1/4^2 + \ldots$ is bounded above by the number 1 and thus by Axiom 2 that the sequence has a limit.

Problem 67. *Show that the limit of the series* $1/2^2+1/3^2+1/4^2+\ldots$ *is less than* 1.

The series $1/2^2+1/3^2+1/4^2+\ldots$ is sometime written as $\sum_{i=2}^{\infty} 1/i^2$. But you must be careful when you see this as books often use the same symbol to mean the limit of the series. For example you might see, if $-1<x<1$, $\sum_{i=0}^{\infty} x^n = \frac{1}{1-x}$ to mean that $\frac{1}{1-x}$ is the limit of the series $1+x+x^2+\ldots$. Note that here and frequently when writing series using the Σ notation, x^0 is used to represent 1. This is a bad use of notation, especially when x can be 0. I prefer to write $1+\sum_{i=1}^{i=\infty} x^n = \frac{1}{1-x}$.

Here is another example of the use of the integral test.

Problem 68. *As was done for the series* $1/2^2+1/3^2+1/4^2+\ldots$ *in Problem 65 show that the series* $1/2^3+1/3^3+1/4^3+\ldots$ *has a limit.*

Problem 69. *For each positive integer n, let* $a_n=\frac{1}{1+n^2}$. *Use the integral test to show that* $a_1+a_2+a_3+\ldots\ldots$ *is a bounded series and thus converges.*

- I suspect I should add more integral test problems. Perhaps as a new classification. Exercises perhaps. Not to be turned in but to be talked about if they can't get them. $\sum \frac{1}{1+i^2}$ for example. They have probably forgotten how to integrate and may have problems with $lim_{x\to\infty} \arctan x$.
  The next problem is for that purpose. If they need that $\sqrt[3]{x}$ is an increasing graph, we can discuss that in class.

Problem 70. *Show that* $\sum_{n=2}^{\infty} \frac{1}{n^3-1}$ *has a limit. Hint: use our rules for inequalities to show that if n is a positive integer and* $n>1$, *then* $\frac{1}{n^3-1} < \frac{2}{x^3}$.

Now we will use the integral test to get part of what is called the **p-test**.

We have shown that if $p=2$ or $p=3$, then the series $1/2^p+1/3^p+1/4^p+\ldots = \sum_{n=2}^{n=\infty} 1/n^p$ has a limit. We now want to show that the same method shows that it has a limit for any number $p>1$.

Problem 71. *Show that if p is a number,* $p>1$, *and* $f(x)=1/x^p$ *for each number* $x>0$ *and n is any positive integer then* $\int_1^n f < \frac{1}{p-1}$. *As always if this seems too hard pick a value for p and work the problem. We have done this for* $p=2$ *and* $p=3$. *Try* $p=1.1$

Problem 72. *Show that if p is a number and* $p>1$, *the series* $1/2^p+1/3^p+1/4^p+\ldots$ *has a limit which is* $\frac{1}{p-1}$ *or less. As before when we had* $p=2$ *or* $p=3$ *show that* $1/2^p+1/3^p+1/4^p+\ldots+1/n^p$ *is a lower Riemann sum for an integral.*

This is part of the p-test. The other part is to show that if $p\le 1$, then the series $1/2^p+1/3^p+1/4^4+\ldots$ does not have a limit.

Definition 18. *The statement that a series or sequence* **diverges** *means it does not converge.*

Thus we have that if an increasing sequence does not have an upper bound, then it diverges.

Problem 73. *First consider the series* $1/2+1/3+1/4+\ldots$ *which is the* $p-$*series when* $p=1$ *Show that if* n *is a positive integer, then* $1/2+1/3+1/4+\ldots+1/n$ *is a lower Riemann sum for* $\int_1^n h$. *Here* $h(x)=1/x$ *for* $x>0$ *as before. Also show that* $1/2+1/3+1/4+\ldots 1/n$ *is an upper Riemann sum for* $\int_2^{n+1} h$. *Explain how this shows that* $\ln(n+1)-\ln 2 < 1/2+1/3+1/4+\ldots+1/n < \ln n$

Problem 74. *Use the proceeding problem and your calculator to find an integer n such that* $1/2+1/3+1/4+\ldots+1/n>10$.

Problem 75. *Show that* $1/2+1/3+1/4+\ldots$ *has no upper bound. That is show that if A is a number then A is not an upper bound by showing that there is an integer n, depending on A so that* $1/2+1/3+1/4+\ldots+1/n>A$

Next we consider the case where $p<1$. And again we will use the integral test. First we consider the case where $p=1/2$.

Problem 76. *Show that* $1/\sqrt{2}+1/\sqrt{3}+1/\sqrt{4}+\cdots+1/\sqrt{n}>\int_2^{n+1}1/\sqrt{x}$.

Problem 77. *Let A be a positive number. Find an integer n such that* $\sum_{k=1}^{k=n}\frac{1}{\sqrt{k}}>A$*, to show that the series diverges.*

Note that in the next problem I have used the dx that you are used to. I use it to describe what function I want to integrate because of the p. I am assuming that p is a constant, and the function I want to integrate is f where $f(x)=1/x^p$. It could have been written as $\int_1^n f$ but that would have required us to define f as was done above. Most books do it as stated.

- Hence the following theorem. If they do not comment about this, I will indicate at some point that for these problems we need that our function is decreasing.

Problem 78. *Assume that* $p<1$*. Show that* $\int_1^n 1/x^p dx=\frac{n^{1-p}-1}{1-p}$*. Now find an integer n, depending on p so that* $\int_1^n 1/x^p dx>10$.

Problem 79. *Assume that* $p<1$ *and that A is a positive number. Find an integer n, depending on A and p, such that* $\int_1^n 1/x^p dx>A$.

There is an easier way to do the preceding problem using what are called comparison tests. The next few problems are about these tests.

Problem 80. *Assume that* $a_1,a_2,a_3,\ldots$ *is a sequence of positive numbers and there is a number A such that for each positive integer n,* $a_1+a_2+a_3+\ldots a_n<A$*, so that the series* $a_1+a_2+a_3+\ldots$ *has a limit. Assume that* $b_1,b_2,,b_3,\ldots$ *is a sequence of positive numbers such that if n is a positive integer,* $b_n\leq a_n$*. Show that for each positive integer n* $b_1+b_2+\ldots+b_n<A$ *so that the series* $b_1+b_2+b_3+\ldots$ *has a limit also.*

Problem 81. *Assume that* $a_1,a_2,a_3,\ldots$ *is a sequence of positive numbers and for each number A there is a positive integer n such that* $a_1+a_2+a_3+\ldots a_n>A$*, so that the series* $a_1+a_2+a_3+\ldots$ *is unbounded and has no limit. Assume that* $b_1,b_2,,b_3,\ldots$ *is a sequence of positive numbers such that if n is a positive integer,* $b_n\geq a_n$*. Show that for each number A, there is a positive integer n such that* $b_1+b_2+\ldots+b_n>A$ *so that the series* $b_1+b_2+b_3+\ldots$ *has no limit also.*

Problem 82. *As an example of how one might use the comparison test, for each positive integer n, let* $b_n=\frac{1}{n^2+1}$*. We know that the series* $1/2^2+1/3^2+1/4^2+\ldots$ *is bounded above by* 1*, so first show that the series* $1/1^2+1/2^2+1/3^2+\ldots$ *is bounded above by* 2*. Then let* $a_n=\frac{1}{n^2}$ *for each positive integer n and show that* $b_n\leq a_n$ *so that* $b_1+b_2+b_3+\ldots$ *converges by Problem 80.*

Problem 83. *Use the comparison test to show that* $\sum_{i=1}^{i=\infty} \frac{1}{i^2-1}$ *is bounded above and thus converges. Hint. Compare the terms with those of the series* $\sum_{i=1}^{i=\infty} \frac{2}{i^2}$.

Problem 84. *Use Definition 13 and the fact that* E *is an increasing function to show that if* a *is a positive number, then* $a^p < a$ *if* $p < 1$ *and* $a^p > a$ *if* $p > 1$.

Problem 85. *Use the comparison test and Problem 75 to do Problem 79.*

Integration by Parts

Suppose that f and g are functions with domain $[a,b]$. You surely recall the product rule for the derivative of their product. It states that $(fg)' = fg' + gf'$. That is if $h(x) = f(x)g(x)$ for each number x in $[a,b]$ then $h'(x) = f(x)g'(x) + g(x)f'(x)$ for each number x in $[a,b]$.

Probably you are familiar with the use of what is basically the same rule in finding antiderivatives or indefinite integrals. Does this look familiar?

$$\int udv = uv - \int vdu$$

Do you realize this is just the product rule in a different form?

We want to use a similar formula for definite Riemann integrals so, starting with $h'(x) = f(x)g'(x) + g(x)f'(x)$ for each number x in the interval $[a,b]$, then we note that:

$$f(x)g'(x) = h'(x) - g(x)f'(x)$$

So we can integrate to get:

$$\int_a^b f(x)g'(x)dx = \int_a^b h'(x)dx - \int_a^b g(x)f'(x)dx$$

or, deleting the $dx's$ for simplicity:

$$\int_a^b fg' = \int_a^b h' - \int_a^b gf'$$

and using Theorem 4, since $h(x) = f(x)g(x)$ we have:

$$\int_a^b fg' = f(b)g(b) - f(a)g(a) - \int_a^b gf'.$$

Problem 86. *Use Theorem 4 to show that* $e = 1 + \int_0^1 E$.

Problem 87. *Starting with the preceding problem, use the integration by parts formula and see what you get. Note that if you want to use the formula on the integral* $\int_0^1 E$*, then you can use any two functions* f *and* g *as long as* $g(x)f'(x) = E(x)$ *for each number* x *in* $[a,b]$*. I hope I don't spoil the problem by suggesting that you let* $f(x) = E(x)$ *and* $g'(x) = 1$.

In the preceding problem, notice that if $g'(x) = 1$, then for any choice of the number c you could let $g(x) = x + c$. If you let $g(x) = x$ in the preceding problem, then do it again, this time letting $g(x) = x - 1$ and see what you get. This is the next problem.

Problem 88. *Start with* $e = 1 + \int_0^1 E$*, and integrate by parts to get* $e = 1 + 1 + \int_0^1 gE$*, where* $g(x) = x - 1$*. One would usually see this written as* $e = 1 + 1 + \int_0^1 e^x(x-1)dx$.

Our next goal is to get a formula that gives the same formula we got before for the Taylor polynomial of degree n based at 0, with the "error" expressed as an integral. That is, we want to find a simple function for g_n for each positive integer n so that:

$$e = 1 + 1 + 1/2! + 1/3! + \ldots + 1/n! + \int_0^1 E \cdot g_n$$

Then we will make an estimate on the value of the integral to see how accurate our approximation is.

Problem 89. *This is a bit tricky. Can you figure out how to repeat what we did in the preceding problem to find a different function for g and take another step and get*

$$e = 1+1+1/2+\int_0^1 Eg\ ?$$

Problem 90. *Do it again to find a function g so that* $e = 1+1+1/2++1/3!+\int_0^1 Eg$?. *Can you see the pattern and guess the formula if we were to do this again?*

Here is the result we have been working toward.

Problem 91. *Show that if n is a positive integer, then*

$$e = 1+1+1/2!+\ldots+1/n!+\int_0^1 \frac{(1-x)^n}{n!}e^x dx.$$

Now we want to use the result of the preceding problem to be able to compute the numerical value of e to any desired degree of accuracy. Note that from the previous problem we have that, if n is a positive integer, then

$$e-(1+1+1/2+\ldots+1/n!) = \int_0^1 \frac{(1-x)^n}{n!}e^x dx$$

Thus the integral in the previous equation is the error in using $1+1+1/2!+\ldots+1/n!$ as an approximation for e. To estimate the error we will use the following fact without proving it. I hope it is easy to believe.

Theorem 5. *If f and g are functions which have Riemann integrals on the interval* $[a,b]$ *and for each number x in* $[a,b]$, $0 \leq f(x) \leq g(x)$, *then* $0 \leq \int_a^b f \leq \int_a^b g$.

Problem 92. *Recall that we have shown (see Problem 38) that* $e < 4$. *Use this and the preceding theorem to show that*

$$\int_0^1 \frac{(1-x)^n}{n!}e^x dx < \frac{4}{(n+1)!}$$

And that

$$e-(1+1+1/2+\ldots+1/n!) \leq 4/(n+1)!$$

In the next problem I picked $\frac{1}{10^{13}}$ in order to see how many terms of our series had to be added to get the 13 places of accuracy that your TI calculator claims to have. Note that if we wanted more that 13 places of accuracy there would be a problem using your calculator to add up the terms in the series.

Problem 93. *Find an integer n (use your calculator) such that*

$$e-(1+1+1/2+\ldots+1/n!) \leq \frac{1}{10^{13}}$$

.

Power series

We began to study power series when we worked on Taylor Polynomials. Recall that we found that the Taylor polynomial approximation for E of degree n was $P_n(x) = 1 + x + x^2/2! + \cdots x^n/n!$ for each number x. Let x be a number. We got the sequence of approximations $1, 1+x, 1+x+x^2/2, \ldots$ for $E(x)$. Written as a series we got the $1+x+x^2/2!+x^3/4!+\ldots$ as our approximations for $E(x)$. And we saw how to get similar series approximations for other functions.

Definition 19. *A power series is just a series of polynomial approximations. Taylor polynomials give one way of finding power series. But any series that can be written as $a_0 + a_1x + a_2x^2 + \ldots$ where each of $x, a_0, a_1, a_2, \ldots$ is a number is a power series.*

You should be familiar from high school with the power series $1+x+x^2+x^3+\ldots$. It is called a geometric series because each term is gotten from the preceding one by multiplying by the same number. For example, if $x = 1/2$, then the series is $1+1/2+1/4+1/8+\ldots$. In high school you probably had the following formula: $a + ar + ar^2 + ar^3 + \ldots = \frac{a}{1-r}$.

Problem 94. *Show that if n is a positive integer, then $\frac{1-x^{n+1}}{1-x} = 1+x+x^2+\ldots+x^n$.*

Problem 95. *Show that if n is a positive integer and $0 < x < 1$, then $1+x+x^2+\ldots+x^n = \frac{1}{1-x} - \frac{x^{n+1}}{1-x} < \frac{1}{1-x}$.*

Note that the preceding problem shows that if $0 < x < 1$ then the series $1+x+x^2+\ldots$ is bounded above by $\frac{1}{1-x}$. If time permits, we may show that the following is correct, but for now we will just assume it is true.

Theorem 6. *If $0 < x < 1$, then the increasing sequence $1, 1+x, 1+x+x^2, \ldots$ has $\frac{1}{1-x}$ as its limit. That is, $\frac{1}{1-x}$ is the smallest number larger than each number in the sequence.*

Note 1. *Actually the preceding theorem is true if $-1 < x < 0$ as well and if time permits we will show this also.*

Problem 96. *Let $a_1, a_2, \ldots$ be a sequence of positive numbers, let r be a number such that $0 < r < 1$. Assume that for each positive integer n $a_{n+1} \leq r \cdot a_n$. Show that for each positive integer n, $a_1 + a_2 + \ldots + a_n \leq \frac{a_1}{1-r}$, so that the series $a_1 + a_2 + a_3 + \ldots$ is bounded above and thus has a limit. Hint: Start by showing that $a_1 + a_2 < a_1(1+r)$. Then show that $a_1 + a_2 + a_3 < a_1(1+r+r^2)$. Now can you show why $a_1 + a_2 + \ldots + a_{n+1} < a_1(1+r+r^2+\ldots+r^n) < \frac{1}{1-r}$?*

- Perhaps the next comment should be a problem. Is it really obvious?

From here on we will just assume that e is the limit of the series $1 + \sum_{i=1}^{i=\infty} \frac{1}{i!}$. And that e-1 is the limit of $\sum_{i=1}^{i=\infty} \frac{1}{i!}$ and $e - 1 - 1/1! - 1/2!$ is the limit of the series $\sum_{i=3}^{i=\infty} \frac{1}{i!}$. etc.

Problem 97. *For each positive integer n let $a_n = 1/n!$. Show that $a_2 \leq \frac{a_1}{2}$, $a_3 \leq \frac{a_2}{2}$ and for each positive integer n $a_{n+1} \leq \frac{a_n}{2}$. Just as in the Problem 96, we have $a_1 + a_2 + \ldots + a_{n+1} < (1 + 1/2 + 1/2^2 + \ldots + 1/2^n) < \frac{1}{1-1/2} = 2$. So $1 + \sum_{i=1}^{i=\infty} \frac{1}{i!}$ is bounded by 3. So we have $e \leq 3$.*

Next we try to improve our approximation for e. I will pick $n = 9$ arbitrarily and use the ratio test to estimate the error in using $\sum_{i=1}^{i=9} \frac{1}{i!}$ as an approximation for e.

Problem 98. *First we want to show that the series* $1/10! + 1/11! + \dots$ *is bounded above. To this end, let* $n > 10$ *be a positive integer and show that* $1/10! + 1/11! + \dots + 1/n! = \frac{1}{10!}(1 + \frac{1}{11} + \frac{1}{11 \cdot 12} + \dots \frac{1}{11 \cdot 12 \cdot \dots n})$.

Problem 99. *Now show that if* $n > 10$ *is a positive integer, then* $\frac{1}{10!}(1 + \frac{1}{11} + \frac{1}{11 \cdot 12} + \dots \frac{1}{11 \cdot 12 \cdot \dots n}) < \frac{1}{10!}(1 + 1/11 + 1/11^2 + \dots + 11^{n-10}) < \frac{1}{10! \cdot (1 - \frac{1}{11})}$

Problem 100. *Is it clear now that we have* $e - (1 + 1 + 1/2! + \dots 1/9!) < \frac{1}{10! \cdot (1 - \frac{1}{11})}$*? Use your calculator to show* $e - (1 + 1 + 1/2! + \dots 1/10!) < .00000031$. *Show that* $1 + 1 + 1/2! + \dots + 1/10! < e < 1 + 1 + 1/2! + \dots + 1/10! + \frac{11}{10 \cdot 10!}$.

We have not discussed what it means to say that a series converges if it is neither increasing nor decreasing, as for example the series $1 - 1/2 + 1/3 - 1/4 + 1/5 \dots$. Intuitively it means that we can compute an approximation to the limit to any desired accuracy, just by adding enough terms. We may not use this definition but for the sake of completeness in your notes, here is the definition.

Definition 20. *If* $a_1, a_2, a_3, \dots$ *is a sequence of numbers, then the statement that* L *is the limit of the sequence* $a_1, a_2, a_3, \dots$ *means that if* c *is a positive number there is a positive integer* n *such that each of* $a_n, a_{n+1}, a_{n+2} + \dots$ *is in the segment* $(L - c, L + c)$.

Problem 101. *We know from Problem 6 that the limit of the series* $1 + 1/2 + 1/4 + 1/8 + \dots$ *is* 2. *Now consider the series* $3 + 2 + 1 + 1/2 + 1/4 + 1/8 + \dots$. *Show that the limit of this series is* 7.

And here is a similar problem which is a bit tricky to show.

Problem 102. *If* $a_1 + a_2 + \dots$ *is a series of positive numbers which has a limit* A, *and* k *is a positive integer, then the series* $a_k + a_{k+1} + a_{k+2} + \dots$ *has* $A - (a_1 + a_2 + \dots + a_k)$ *as a limit*

The next theorem we will not take the time to prove but it will help explain why I have spent so much time on series where all the terms are positive.

Theorem 7. *If* $a_1 + a_2 + a_3 + \dots$ *is a series of numbers, not necessarily all positive then the series* $a_1 + a_2 + \dots$ *has a limit if the series* $|a_1| + |a_2| + \dots$ *has a limit.*

Note 2. *In the case of power series, the following is true. It will explain in part why I concentrated on only sequences of positive terms. We will not show this to be true, but you may well need this in later courses.*

As an example of a series for which the preceding theorem does not apply, recall that $1 + 1/2 + 1/3 + \dots$ is not a bounded series and thus does not have a limit. However the series $1 - 1/2 + 1/3 + 1/4 - 1/5 + \dots$ does have a limit. This is easy to believe from the next problem.

Problem 103. *For each positive integer* n, *let* $a_n = 1 - 1/2 + 1/3 - 1/4 + \dots + \frac{(-1)^{n+1}}{a_n}$. *Show that* $a_1, a_3, \dots$ *is a decreasing bounded sequence and* $a_2, a_4, \dots$ *is an increasing bounded sequence and they have the same limit.*

Theorem 8. *If $a_0, a_1, a_2, \ldots$ is a sequence of numbers then*
1) $a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots$ converges only if $x = 0$, or
2) $a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots$ converges for every number x, or
3) there is a number R such that the $a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots$ converges if $-R < x < R$ and does not converge if $x < -R$ or $x > R$.
In the last case the series might or might not converge if $x = R$ or $x = -R$.

Here is the way the ratio test is usually stated for power series. Again we will not prove this.

Theorem 9. *If $a_0 + a_1x + a_2x^2 + \ldots$ is a power series then the series converges if the sequence $\frac{|a_1x|}{|a_0|}, \frac{|a_2x|}{|a_1|}, \frac{|a_3x|}{|a_2|}, \ldots$ has a limit which is less than 1. And the power series does not converge if the limit of this sequence is more then 1.*

As an example of the use of Theorem 9 we return to our Taylor's series for $f(x) = e^x$.

Problem 104. *Consider the series $1 + x + x^2/2! + x^3/3! + \ldots$. Now form the sequence of ratios of consecutive terms of this series. To avoid the use of absolute values, assume that x is positive. The ratios are $\frac{x}{1}, \frac{x^2/2!}{x}, \frac{x^3/3!}{x/2!}, \ldots$. Show that these have 0 as a limit. Thus the series converges by Theorem 9 for any choice of x.*

Problem 105. *Next consider the series $1 + x + x^2 + \ldots$. As before find the sequence which is made up of ratios of consecutive terms. What would you say is the limit of this sequence? One of the facts about sequences is that a sequence which repeats the same number forever has a limit which is that number. So the limit of our sequence is x. If $|x| < 1$, then then series converges and if $|x| > 1$ it does not converge by Theorem 9. So we know that the series converges for $-1 < x < 1$, and it does not converge if $x > 1$ or $x < -1$. The ratio test does not help in case $x = 1$ or $x = -1$. To see what happens in this case, return to the original series and substitute 1 for x and then -1 for x and look at the series of partial sums $s_1, s_2, \ldots$ where for each positive integer n, $s_n = 1 + x + x^2 + \ldots + x^n$.*

Chapter 6

Comments and student feedback

I have taught this course 5 times. The first time I thought the course was too unstructured and abstract. And the next time I thought I had too much structure. After that I just let the students decide how abstract it should be. But the course was very well received by the students even when I thought I had done a bad job. Apparently knowing in advance that it was to be a seminar course without lectures made for a select group of dedicated students who enjoyed the challenges.

There seems to be a general perception that this type of course causes a split in the students with some liking the course very much and getting a lot from it while others dislike it and benefit little. This has not been my experience.

The administration requires that the faculty hand out a standard evaluation form in each class. The questions are divided into two groups. One groups asks questions about the course and the other group asks questions about the instructor. The students answer the questions on a scale of 1-9. The scores I and the course have received are as follows:

Semester	Course	Instructor
Fall 1998	6.29	7.73
Fall 1999	6.74	7.87
Spring 2000	6.07	7.05
Spring 2001	8.45	8.17
Fall 2001	6.71	7.76

I find it quite impossible to predict what score the students will give me or the course.

In addition to these figures, the mathematics department requires the faculty to hand out a sheet which asks the students to respond to the following statements.

1. Please comment on the strengths and weaknesses of

A. The Course

B. The Instructor

2. Would you recommend this course or this instructor to another student? Why?

Of 50 written responses, none were strongly unfavorable and only 8 made any comment that was not complementary. These comments generally expressed a concern that not enough material had been covered or that not enough routine problems had been given, or that the course was too theoretical.

I will include below a few selected comments that I think tell something about the course from the other 48 responses.

Learning to derive formulas instead of just having them thrown at you was definitely a strength, having classmates work problems and try to figure solutions out together helped a great deal.

goes through many topics, and teaches the student how to approach unknown problems.

It was fun to learn how the laws of calculus were developed. Each problem led to an endpoint that was withheld until the right time. Students were encouraged to guess why we were doing certain problems. It was fun too.

Course made students think in a new way, try things we never had before, attempt problems we thought would be impossible because we saw them in a new light.

This course encouraged students to think for themselves instead of just following instructions. The course structure allowed for a lot of self paced work.

He challenged us to figure out problems I never thought I would be able to figure out.

Each year, our department teaches 4-6 sections totaling over 100 students of a special section of our 2^{nd} semester calculus for students who made a 4 or 5 on the AB Calculus advanced placement exam. These students receive a letter from the department inviting them to take this course without first taking the usual 1^{st} semester calculus for which they receive AP credit. Each section has about 25 students. My course has the same requirements and is for the same students, but the maximum class size is 15 students. No records have been kept but the general feeling is that we attract few of these AP students into mathematics. Indeed I suspect that few of them even take additional math courses. When I taught my seminar in the Fall of 2001 I decided to track my students. While this is a small sample, I expect to continue this in the future. There were 11 students in the class. Of those 11, 4 took another mathematics course the next semester, one of those plans to major in mathematics. Of the other 7, one said they planned to take more mathematics and one said they were considering a minor in mathematics.